



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 6

- Case Study Title:** To demonstrate JavaScript string operations using `substring()`, `indexOf()`, `split()`, `replace()`, string reversal, and vowel counting based on user input.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>String Operations</title>
  <style>
    body {
      background-color: rgb(239, 171, 243);
      font-family: Arial;
    }
    input {
      padding: 8px;
      margin: 5px;
      width: 350px;
    }
    textarea {
      padding: 10px;
      margin: 5px;
      width: 500px;
      height: 80px;
    }
    button {
      padding: 10px 20px;
      margin: 10px 5px;
    }
    #output {
      margin-top: 20px;
      font-size: 18px;
    }
  </style>
```



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```
</head>
<body>
  <h2>Shreya's JavaScript String Operations</h2>
  <label>Enter String:</label><br>
  <textarea id="str" placeholder="Enter your string or
paragraph"></textarea>
  <br>
  <label>Substring Start:</label>
  <input type="number" id="start" placeholder="Enter start index">
  <br>
  <label>Substring End:</label>
  <input type="number" id="end" placeholder="Enter end index">
  <br>
  <label>IndexOf - Enter text to search:</label>
  <input type="text" id="search" placeholder="Enter character or
word">
  <br>
  <label>Split - Enter separator:</label>
  <input type="text" id="separator" placeholder="Example: space">
  <br>
  <label>Replace - Text to replace:</label>
  <input type="text" id="oldText" placeholder="Enter text to replace">
  <br>
  <label>Replace - New text:</label>
  <input type="text" id="newText" placeholder="Enter new text">
  <br>
  <button onclick="performOperations()">Perform Operations</button>
  <div id="output"></div>
  <script src="shreya_case6.js"></script>
</body>
</html>

function performOperations() {

  let str = document.getElementById("str").value;

  let start = Number(document.getElementById("start").value);
  let end = Number(document.getElementById("end").value);

  let search = document.getElementById("search").value;
  let separator = document.getElementById("separator").value;
```



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```
let oldText = document.getElementById("oldText").value;
let newText = document.getElementById("newText").value;
let output = "";
// Substring
output += "Substring: " + str.substring(start, end) + "<br>";
// IndexOf
output += "Index of '" + search + "': " + str.indexOf(search) +
"<br>";
// Split
if (separator === "space") {
    output += "Split: " + str.split(" ") + "<br>";
} else {
    output += "Split: " + str.split(separator) + "<br>";
}
// Replace
output += "Replace: " + str.replace(oldText, newText) + "<br>";

// Reverse
let reverse = str.split("").reverse().join("");
output += "Reversed String: " + reverse + "<br>";
// Count Vowels
let count = 0;
for (let i = 0; i < str.length; i++) {
    let ch = str[i].toLowerCase();
    if (ch == "a" || ch == "e" || ch == "i" ||
        ch == "o" || ch == "u") {
        count++;
    }
}
output += "Number of Vowels: " + count;
document.getElementById("output").innerHTML = output;
}
```



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3. Output:

Fwd: js1ab 7 - shreya.maske.b

String Operations

File /Users/shreyamaske/javascript_exp/jsexp6/shreya_case6.html

Shreya's JavaScript String Operations

Enter String:

My name is Shreya Maske.I love Javascript.
Studying in SIT Nagpur.
24070521196

Substring Start:

Substring End:

IndexOf - Enter text to search:

Split - Enter separator:

Replace - Text to replace:

Replace - New text:

Perform Operations

Substring: me is Shreya Maske.I love Javascript. Studying in SIT Nagpur. 24070521
Index of 'SIT': 55
Split: My name is Shreya Maske,I love Javascript, Studying in SIT Nagpur, 24070521196
Replace: My name is Shreya Maske.I like Javascript. Studying in SIT Nagpur. 24070521196
Reversed String: 69112507042 .rupgaN TIS ni gniydutS .tpircsavaJ evol l.eksaM ayerhS si eman yM
Number of Vowels: 19

4. Result/Conclusion:

Thus, various JavaScript string operations were successfully performed, including **substring extraction**, **text searching**, **splitting**, **replacing**, **reversing**, and **vowel counting**, and the results were displayed dynamically on the webpage.